

EL MEHDI HICHAM

Computer Vision Engineer | AI Engineer | Sports Analytics Morocco — open to relocation & remote

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PROFILE

Computer Vision Engineer specialized in **real-time sports analytics** and **multi-object tracking**. Builder of **TactiVision**, an independent R&D project turning broadcast video into structured tactical data using TensorRT, CUDA Streams, homography, pose estimation, OCR and re-identification. Previously contributed to AI systems at SportScore and Priori Sports, with a strong focus on scalable, production-ready computer vision pipelines.

EXPERIENCE

Computer Vision Engineer (Contract) 05/2026 – 06/2026
Priori Sports, Inc. — San Francisco, USA (Remote)

- Developed AI and computer vision solutions for **football spatial tracking**, including homography-based projection systems and tracking pipeline optimization.
- Built **ball trajectory modeling** and airborne ball localization using tracking data and spatial correction techniques.
- Contributed to real-time sports analytics workflows and visualization tools for performance analysis.

Tech: Python, PyTorch, OpenCV, Homography, CUDA, TensorRT, Docker

TactiVision — Independent R&D Project 01/2026 – Present
Football video analysis pipeline mehdihch.github.io/PortfolioMCH

- Built and deployed an end-to-end CV pipeline running **7 TensorRT engines in parallel** via CUDA streams on a single RTX 4060: player/GK/referee detection, ball detection, pitch keypoint homography, 17-point pose estimation, field segmentation, jersey OCR, and OSNet re-identification.
- Dedicated goalkeeper model** outputting GK position in real-world metres, speed, skeleton pose and spatial goal coverage on every frame.
- Trained on a **140,000+ image dataset** assembled across 7 competitions (LaLiga, Premier League, Ligue 1, Bundesliga, UCL, Botola Pro, FIFA World Cup 2026), reaching **93.9–98.2% detection precision** on held-out footage from unseen matches.
- Match-centre dashboard generating possession, territory, xThreat, pass networks, heatmaps, pitch control and pressing intensity automatically from a **single broadcast feed** — no stadium infrastructure required.
- Every output ships with **measured coverage and confidence** (tracking coverage, calibration status, ball visibility, share of time under confirmed identity) rather than unqualified statistics.

Tech: TensorRT, ONNX, CUDA Streams, Ultralytics YOLO, ByteTrack, OSNet, OpenCV, PyTorch

R&D Intern — Applied AI & Computer Vision 04/2025 – 10/2025
SportScore, Paris (Remote)

- Built an automatic frame classification module (YOLOv11m-cls) achieving **99.95% accuracy**; implemented complex action detection (pass classification, shots, dribbles, interceptions, goals).
- Optimized the vision pipeline end-to-end: **50% latency reduction** via CUDA Stream parallelization (5 streams) and TensorRT FP16 (7 models), achieving **13 FPS (+117%)**.
- Applied camera homography and geometric transformations for spatial analysis and field mapping.
- Delivered automated performance analytics (heatmaps, pass maps, structured JSON exports, match statistics) and **AI-powered microservices** integrated into production systems.
- Worked autonomously and **mentored 4 interns**, ensuring code quality and consistency.

Tech: Python, PyTorch, TensorFlow, OpenCV, CUDA, TensorRT, Pandas, Docker, FastAPI

Machine Learning Intern — Research & Experimental Evaluation 10/2024 – 01/2025
SETIME Lab, Ibn Tofail University

- Designed reproducible ML pipelines on a **40GB dataset** (2024 Futsal World Cup, 7 matches); conducted controlled experiments with benchmarking and ablation studies.
- Achieved **0.93 recall** using RegNet-Y + Bi-GRU; performed hyperparameter tuning and architecture exploration.

Tech: Python, PyTorch, RegNet-Y, ResNet-50, Bi-GRU, OpenCV, NumPy, Pandas

PROJECTS

OCR System for Handwritten Tifnagh 2024
Hybrid CRNN (CNN + RNN) for Amazigh script recognition; benchmarked against PaddleOCR.

Autonomous Driving — Deep Reinforcement Learning 2024
ROS2 + Gazebo simulation pipeline (camera, LiDAR); TD3 agent for navigation and obstacle avoidance.

SKILLS

Computer Vision: YOLOv8/11, ByteTrack, DeepSORT, OSNet ReID, OCR (PaddleOCR, CRNN), pose estimation, segmentation, homography, camera calibration, 3D geometry

Performance & Deployment: CUDA Streams, TensorRT FP16/INT8, ONNX, real-time inference, Docker, AWS (S3, EC2)

Programming: Python (advanced), C/C++, Shell, SQL, FastAPI, REST API

Deep Learning & ML: PyTorch, TensorFlow/Keras, CNN, RNN/LSTM, Transformers, EfficientNet, RegNet-Y, Bi-GRU

Mathematics: linear algebra, projective geometry, statistical analysis, signal processing, numerical methods

Robotics & Tools: ROS2, Gazebo, LiDAR, camera integration, Pandas, NumPy, Git, Linux

Languages: Arabic (native), French (professional), English (professional working proficiency)

EDUCATION

Master's Degree in Artificial Intelligence — Ibn Tofail University, Kenitra 2023 – 2025

Bachelor's Degree in Physics (Theoretical & Mathematical) — Ibn Tofail University, Kenitra 2020 – 2023